Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

Question: Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

Answer:

```
DECLARE
  CURSOR cust_cursor IS
    SELECT customer_id, age, loan_interest_rate
    FROM customers;
BEGIN
  FOR rec IN cust_cursor LOOP
    IF rec.age > 60 THEN
      UPDATE customers
      SET loan_interest_rate = loan_interest_rate - 0.01
      WHERE customer_id = rec.customer_id;
    END IF;
  END LOOP;
  COMMIT;
END;
/
```

Scenario 2: A customer can be promoted to VIP status based on their balance.

Question: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over $10,000.

Answer:

```
DECLARE
  CURSOR cust_cursor IS
```

```
        SELECT customer_id, balance
        FROM customers;
BEGIN
    FOR rec IN cust_cursor LOOP
        IF rec.balance > 10000 THEN
            UPDATE customers
            SET IsVIP = 'TRUE'
            WHERE customer_id = rec.customer_id;
        END IF;
    END LOOP;
    COMMIT;
END;
/
```

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

```
DECLARE
    CURSOR loan_cursor IS
        SELECT customer_id, loan_id, due_date
        FROM loans
        WHERE due_date BETWEEN SYSDATE AND SYSDATE + 30;
BEGIN
    FOR rec IN loan_cursor LOOP
        DBMS_OUTPUT.PUT_LINE('Reminder: Loan ' || rec.loan_id ||
                             ' for Customer ' || rec.customer_id ||
                             ' is due on ' || TO_CHAR(rec.due_date, 'DD-MON-YYYY'));
```

```
   END LOOP;
END;
/
```